

Unit 10: NLP Reflection

1. What?

In this unit, I learned about **Natural Language Processing (NLP)** and the evolution from traditional NLP approaches to modern transformer-based models.

Key concepts included:

- **Transformer architecture** and its advantages over Recurrent Neural Networks (RNNs), particularly its ability to process data in parallel, which significantly improves efficiency and scalability.
- **Multi-head attention**, which allows the model to focus on different parts of a sentence simultaneously and capture complex contextual relationships.
- **Tokenization** and **word embeddings**, which convert text into numerical representations that machine learning models can understand.
- **GPT (Generative Pre-trained Transformer)** models, which are based on **decoder-only transformers** and represent a major advancement in modern NLP.
- The differences between **classical NLP approaches** and **modern NLP systems**, highlighting how transformer-based architectures have transformed the field.

2. So What?

This unit was particularly interesting because it allowed me to revisit **sentiment analysis**, which I had previously explored in a team project. Reflecting on that experience helped me better understand how NLP techniques have evolved over time.

In our project:

- We used an **RNN-based architecture** rather than a transformer.
- Similar to modern NLP systems, we still relied on **tokenization** and **word embeddings** to represent text data.
- The project required effective collaboration among team members and gave me practical experience in applying NLP concepts to a real-world problem.

Comparing our sentiment analysis model with modern GPT-based systems helped me appreciate the significant improvements brought by transformer architectures, particularly in handling context, scalability, and overall performance.

3. Learning, Change, and Future Application

A key finding from the e-portfolio activity, where I implemented a GPT-like sentiment analysis model and compared it with a classical RNN-based approach, was that a more expressive architecture does not necessarily lead to better performance. Since both models were trained on a relatively limited dataset, the transformer-based model showed little improvement over the RNN. This highlighted that model performance depends not only on architectural complexity but also on the quality and quantity of the available training data.

This unit significantly enhanced my understanding of modern Natural Language Processing and the technologies that power today's AI systems. While I was already familiar with NLP concepts through previous sentiment analysis projects, learning about transformer architectures provided a deeper understanding of how large language models such as GPT achieve superior performance compared to traditional RNN-based models. I gained a clearer appreciation of the role of mechanisms such as attention and parallel processing in improving both efficiency and the ability to capture contextual relationships within text.

The most important change in my learning was the shift from viewing NLP as primarily a text classification tool to recognizing it as a foundation for advanced AI applications, including conversational agents, content generation, translation, summarization, and intelligent decision-support systems. Understanding the evolution from RNNs to transformers has given me a broader perspective on how rapidly the field is advancing and reinforced the importance of continuously updating my technical knowledge to remain relevant in an AI-driven environment.

This learning will be valuable in my future academic and professional work. I can now better evaluate the strengths and limitations of different NLP approaches and make more informed decisions when selecting models for specific tasks. In future projects, I would like to experiment with transformer-based models for sentiment analysis and compare their performance with traditional RNN approaches to better understand the practical benefits of modern architectures. More broadly, the knowledge gained from this unit has increased my confidence in working with AI technologies and will help me contribute more effectively to projects involving language data, machine learning, and intelligent systems.